CHAPTER 1

INTRODUCTION

1.1 Overview of Vibrio vulnificus

Vibrio vulnificus is a Gram-negative, halophilic bacterium that inhabits coastal and estuarine marine environments. It is an opportunistic human pathogen primarily transmitted through the consumption of raw or undercooked seafood, particularly oysters, or through exposure of open wounds to contaminated seawater. Clinically, V. vulnificus causes three principal disease manifestations: primary septicaemia, wound infection, and gastroenteritis. The bacterium's remarkable growth rate at warm temperatures and its proficiency in evading host immune defenses render it particularly dangerous in immunocompromised individuals, in whom case fatality rates can exceed 50%.

Global warming and the consequent rise in coastal water temperatures have expanded the ecological range of V. vulnificus poleward, increasing the population at risk of infection. Concurrently, intensified aquaculture practices have created dense, antibiotic-laden environments that may accelerate the selection and dissemination of resistant strains, compounding the clinical management challenge.

1.2 Virulence and Antimicrobial Resistance

The pathogenicity of V. vulnificus is multifactorial, underpinned by an arsenal of virulence determinants that collectively enable host colonization, immune evasion, and tissue destruction. Capsular polysaccharides (CPS) mask the organism from phagocytosis and complement-mediated killing. Siderophores scavenge iron from host proteins, satisfying the organism's requirement for this cofactor during infection. A pore-forming cytolysin (VvhA) disrupts erythrocyte and leukocyte membranes, releasing nutrients and triggering inflammatory cascades. Additional virulence factors include the multifunctional autoprocessing repeats-in-toxin (MARTX) toxin, collagenase, and a type III secretion system.

Against this virulence backdrop, the emergence of antimicrobial resistance (AMR) in V. vulnificus threatens to erode the efficacy of the antibiotics currently relied upon for treatment. Fluoroquinolones (ciprofloxacin) and third-generation cephalosporins, often administered in combination, constitute the standard-of-care regimen. However, resistance mechanisms targeting these drug classes — including mutations in DNA gyrase and topoisomerase IV, beta-lactamase production, and efflux pump upregulation — have been documented with increasing frequency.

1.3 Marine-Derived Compounds and Manoalide

Marine organisms represent a largely untapped reservoir of structurally diverse and biologically active secondary metabolites. Sponges, corals, tunicates, cyanobacteria, and microalgae synthesize compounds evolved for predator deterrence, competition for space, and microbial defense — a selective context that mirrors many of the mechanisms clinically relevant pathogens employ. Among these marine-derived molecules, Manoalide stands out for its potent and well-characterized bioactivity.

Manoalide (PubChem CID: 6437368; molecular formula $C_{25}H_{36}O_5$; MW 416.55 g/mol) is a sesterterpenoid sesquiterpene isolated from the marine sponge Luffariella variabilis. Its primary documented mechanism is irreversible inhibition of phospholipase A_2 (PLA_2) through covalent modification of active-site lysine residues, a property that underpins its potent anti-inflammatory activity. Manoalide entered clinical trials as a topical anti-inflammatory agent in the 1990s and, although development was ultimately discontinued for commercial reasons, its clinical-stage history provides reassurance regarding its tolerance in biological systems.

Beyond PLA_2 inhibition, Manoalide has demonstrated activity against a range of enzymatic and protein targets in vitro, suggesting that its bifunctional butenolide and γ-hydroxybutenolide ring system can engage multiple protein scaffolds. This chemical promiscuity, if directed against microbial resistance determinants, could yield a novel antibiotic adjuvant with a high barrier to resistance.

1.4 Bioinformatics and Computational Approaches

The rapid accumulation of microbial genome sequences and protein structural data, combined with advances in computational chemistry, has transformed the early stages of drug discovery. In silico methodologies now enable researchers to prioritize lead compounds from vast chemical libraries before any bench experiment is performed, substantially reducing time and cost.

Genome-scale gene prediction tools such as Prodigal (Prokaryotic Dynamic Programming Genefinding Algorithm) generate comprehensive catalogues of protein-coding sequences from assembled microbial genomes. Resistance gene databases — most notably the Comprehensive Antibiotic Resistance Database (CARD), interrogated via the Resistance Gene Identifier (RGI) tool — then link predicted protein sequences to curated resistance determinants, enabling rapid identification of AMR gene content.

Molecular docking algorithms model the non-covalent interaction between a small-molecule ligand and a protein target to estimate binding affinity and predict binding geometry. The CB-Dock2 platform, which integrates cavity detection, AutoDock Vina–based docking, and homologous template fitting, provides both affinity estimates and structural context for ligand–

protein complexes. SwissADME further contextualizes docking hits by predicting absorption, distribution, metabolism, and excretion (ADME) properties, including Lipinski's Rule of Five compliance, which is a widely accepted filter for estimating oral bioavailability.

CHAPTER 2

AIM AND OBJECTIVES

2.1 Aim

To evaluate the antimicrobial potential of Manoalide, a marine-derived sesterterpenoid, against antimicrobial resistance (AMR) determinants of Vibrio vulnificus through an integrated in silico approach encompassing genome annotation, resistance gene identification, molecular docking, and pharmacokinetic profiling.

2.2 Objectives

1. To retrieve and process the complete genome sequence of Vibrio vulnificus from the NCBI Genome Database.
2. To perform gene prediction using Prodigal and identify putative protein-coding sequences.
3. To identify antimicrobial resistance (AMR) genes using the Comprehensive Antibiotic Resistance Database (CARD) and Resistance Gene Identifier (RGI).
4. To select and prepare three-dimensional protein structures of identified AMR targets for molecular docking studies.
5. To retrieve the three-dimensional structure of Manoalide from the CMNPD and PubChem databases and prepare it for docking.
6. To perform molecular docking using CB-Dock2 to determine binding affinities of Manoalide against each identified AMR protein.
7. To analyze and visualize protein–ligand interaction patterns using PyMOL.
8. To evaluate the drug-likeness and pharmacokinetic profile of Manoalide using SwissADME.

CHAPTER 3

REVIEW OF LITERATURE

3.1 Vibrio vulnificus and Its Capacity to Induce Disease

Vibrio vulnificus has been the subject of sustained clinical and microbiological investigation since its formal recognition as a distinct species. Horseman and Surani (2011) provided a comprehensive synthesis of its pathophysiology, epidemiology, and clinical management, establishing it as an important cause of primary septicaemia and necrotizing fasciitis, particularly in patients with liver disease, haematological malignancies, or other immunosuppressive conditions.

The molecular determinants of V. vulnificus virulence have been extensively dissected. Gulig et al. (2003) mapped the genetic architecture of pathogenicity in this organism, identifying the central roles of capsular polysaccharides in complement resistance and opsonization evasion, as well as the contribution of iron acquisition systems to systemic infection. Oliver (2005) further contextualized wound infections caused by V. vulnificus, noting that direct percutaneous inoculation during seafood handling or recreational water exposure constitutes a distinct infection route from foodborne septicaemia, with its own epidemiological profile and clinical course.

Genomic analyses conducted in subsequent years revealed remarkable metabolic plasticity in V. vulnificus. The organism's approximately 5 Mb genome encodes a broad complement of regulatory systems, environmental sensing modules, and metabolic pathways that collectively enable survival and pathogenicity across diverse ecological niches — from oyster hemocytes to human vasculature.

3.2 Antimicrobial Resistance in Marine Ecosystems

Marine environments have emerged as globally important reservoirs and dissemination hubs for antimicrobial resistance determinants. Cabello et al. (2013) demonstrated that aquaculture operations, which employ large quantities of therapeutic and prophylactic antibiotics, function as hotspots for resistance selection and horizontal gene transfer. The proximity of these operations to natural coastal waters facilitates the bidirectional exchange of resistance genes between aquaculture-associated bacteria and wild marine microbial communities.

Miranda and Zemelman (2002) identified substantial multiresistance in bacteria isolated from freshwater aquaculture systems, presaging subsequent reports of similar resistance profiles in marine-adapted organisms. More recent investigations have broadened this picture: Zhang et al. (2025) catalogued the spectrum of AMR genes circulating in marine environments and

characterized the environmental drivers — including antibiotic contamination from pharmaceutical manufacturing discharge, agricultural runoff, and municipal wastewater — that shape resistance selection pressure in marine ecosystems.

Kumar et al. (2022) reviewed the mechanistic diversity of resistance in marine microorganisms, documenting the full spectrum of known resistance strategies in marine-associated bacteria: enzymatic drug inactivation (beta-lactamases, aminoglycoside-modifying enzymes), efflux pump-mediated extrusion, target-site modification (ribosomal mutations, altered penicillin-binding proteins), and target protection. Singh et al. (2025) further emphasized emerging trends in marine AMR, including the increasing prevalence of multidrug-resistant phenotypes and the dissemination of resistance via mobile genetic elements. Li et al. (2025) connected environmental pollution directly to resistance spread, providing mechanistic evidence that anthropogenic chemical stress selects for cross-resistance to antibiotics.

3.3 Genome Mining and Gene Prediction

The development of accurate, computationally efficient gene prediction tools has been transformative for microbial genomics. Prodigal (Prokaryotic Dynamic Programming Genefinding Algorithm), described by Hyatt et al. (2010), represents a landmark advance in prokaryotic gene calling. By integrating dynamic programming with probabilistic models trained on sequence-specific codon usage patterns and translation initiation signals, Prodigal achieves gene-prediction accuracy that rivals manual expert annotation while operating at genome-scale throughput.

Prodigal operates in a fully unsupervised mode, requiring no reference genome or manually curated training set: it derives organism-specific models directly from the input sequence. This self-training architecture makes it particularly well suited to the analysis of novel or divergent microbial genomes for which reference data may be limited. For V. vulnificus, a bacterium with recognized genomic plasticity and strain-to-strain variation, this property is especially valuable.

Genome mining approaches that integrate gene prediction with functional annotation have increasingly been applied to the discovery of novel biosynthetic gene clusters encoding potentially bioactive secondary metabolites. In bacteria such as V. vulnificus, where genomic flexibility may extend to the incorporation and loss of horizontally transferred resistance islands, comprehensive gene prediction followed by database-guided functional assignment provides the most complete picture of encoded resistance potential.

3.4 Utilizing the CARD Database for AMR Gene Identification

The Comprehensive Antibiotic Resistance Database (CARD) was established to provide a curated, regularly updated repository of resistance gene sequences, resistance-conferring mutations, and associated mechanistic information. Jia et al. (2017) described CARD's

foundational architecture, which organizes resistance information around the Antibiotic Resistance Ontology (ARO) — a hierarchical classification system that encodes relationships between drugs, drug classes, resistance mechanisms, and the genes or mutations responsible for each phenotype.

The Resistance Gene Identifier (RGI) tool, which queries protein or nucleotide sequences against CARD's curated models, provides three tiers of evidence-filtered output: 'perfect' hits (100% identity to a curated reference), 'strict' hits (meeting defined similarity thresholds with strong evidence support), and 'loose' hits (lower-confidence predictions that may warrant further investigation). Alcock et al. (2022) described updates to CARD's surveillance capability, incorporating expanded genomic surveillance data and refined classification of resistance gene families.

For organisms such as V. vulnificus, whose AMR gene content may include both intrinsic (chromosomally encoded) and acquired (horizontally transferred) resistance determinants, CARD-RGI analysis provides the systematic framework necessary to distinguish these categories and prioritize the most clinically relevant targets for downstream investigation.

3.5 Molecular Docking in the Search for New Drugs

Molecular docking has become a cornerstone of structure-based drug discovery, enabling the computational prediction of how a small molecule (ligand) binds to a macromolecular target (typically a protein) and with what affinity. Liu et al. (2022) described CB-Dock2, an advanced docking platform that addresses a key limitation of conventional docking: the requirement for a predefined binding site. CB-Dock2 employs CavityPlus for de novo cavity detection, identifying all geometrically feasible binding pockets on the protein surface before initiating AutoDock Vina-based docking calculations in each pocket. This blind-docking capability is particularly valuable when the functional binding site of a resistance protein is unknown or when off-site binding may contribute to inhibitory activity.

The validation of docking predictions against crystallographic or NMR-determined binding poses has confirmed that AutoDock Vina-based methods reliably reproduce experimental binding geometries for a broad range of protein-ligand systems, particularly when binding energies are in the range that is relevant for drug development (−7 to −12 kcal/mol). For natural product scaffolds, which often possess complex three-dimensional shapes and multiple functional groups capable of engaging diverse protein environments, docking scores must be interpreted alongside structural analysis of the predicted binding mode.

3.6 Lipinski's Rule of Five and ADME Analysis

The Lipinski Rule of Five, formulated by Lipinski (2004) based on analysis of orally active drugs in the World Drug Index, establishes empirical boundaries for molecular properties

associated with oral bioavailability: molecular weight ≤ 500 Da, calculated lipophilicity (logP) ≤ 5, hydrogen bond donor count ≤ 5, and hydrogen bond acceptor count ≤ 10. Compounds satisfying all four criteria are considered 'drug-like' in the sense of possessing physicochemical properties conducive to passive gastrointestinal absorption.

Daina et al. (2017) described SwissADME, a freely accessible web tool that extends Lipinski assessment to a comprehensive ADME profile. Beyond Rule of Five compliance, SwissADME reports predicted gastrointestinal absorption probability, blood–brain barrier permeability, P-glycoprotein substrate/inhibitor status, metabolic stability, and bioavailability score. For infectious disease drug candidates, gastrointestinal absorption and metabolic stability are particularly critical parameters, as systemic exposure must be adequate to achieve therapeutically relevant concentrations at the site of infection.

Mayer et al. (2010) documented the broad pharmaceutical pipeline of marine natural products, noting that the structural novelty and biological potency of marine-derived compounds often confer favorable ADME profiles despite their molecular complexity. Blunt et al. (2018) catalogued the continuing contribution of marine natural product chemistry to drug discovery, reinforcing the importance of ADME-guided prioritization in marine compound development.

3.7 Global Burden of Antimicrobial Resistance

Antimicrobial resistance (AMR) has emerged as one of the most pressing global health challenges of the 21st century. The increasing inability of antibiotics to effectively treat infections has resulted in prolonged illness, higher healthcare costs, and increased mortality rates. According to recent global estimates, AMR is responsible for millions of deaths annually, either directly or indirectly, due to treatment failure.

The problem is particularly severe in developing countries, where antibiotic misuse, lack of regulation, and limited healthcare infrastructure accelerate resistance development. In marine environments, the issue is further complicated by environmental pollution, aquaculture practices, and climate change, all of which contribute to the spread of resistant bacteria.

Vibrio species, including Vibrio vulnificus, are increasingly recognized as important contributors to this global AMR crisis. Their natural habitat in marine ecosystems exposes them to diverse microbial communities and environmental stressors, promoting genetic exchange and resistance evolution.

This highlights the urgent need for alternative therapeutic strategies, such as marine-derived bioactive compounds, which may offer novel mechanisms of action against resistant pathogens.

CHAPTER 4

MATERIALS AND METHODS

4.1 Choosing the Target Organism

Vibrio vulnificus was selected as the target organism for this study based on its clinical relevance, increasing antimicrobial resistance (AMR) profile, and the availability of a complete and well-annotated reference genome. The organism is a Gram-negative, halophilic marine bacterium commonly found in coastal and estuarine environments. Its genome provides a suitable model for computational analysis due to its relatively well-characterized genetic composition and the presence of multiple resistance-associated genes.The selection of this organism was further supported by the accessibility of its genomic data through publicly available databases such as the National Center for Biotechnology Information (NCBI). The availability of high-quality genome sequences ensures reliability in downstream bioinformatics analyses, including gene prediction and resistance gene identification. Additionally, the organism's genome size and structure are appropriate for in silico workflows, allowing efficient processing using standard computational tools.From a methodological perspective, Vibrio vulnificus serves as an ideal candidate for studying antimicrobial resistance mechanisms using computational approaches. The presence of diverse resistance genes enables comprehensive analysis through databases such as the Comprehensive Antibiotic Resistance Database (CARD). This facilitates accurate identification and classification of resistance determinants using tools like the Resistance Gene Identifier (RGI).

Furthermore, the availability of experimentally determined or predicted three-dimensional protein structures for several of its resistance-associated proteins allows effective implementation of molecular docking studies. These structural data are essential for evaluating protein–ligand interactions and predicting binding affinities of selected compounds.

Overall, the selection of Vibrio vulnificus provides a robust and reliable model system for conducting genome-based analysis, resistance gene identification, and structure-based drug discovery within a computational framework.

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4.2 Genome Sequence Retrieval and Preparation

The complete genome sequence of *Vibrio vulnificus* was retrieved from the National Center for Biotechnology Information (NCBI) Genome Database in FASTA format. A fully assembled and annotated reference genome was selected to ensure high reliability for downstream computational analyses. The genome, with an approximate size of 5.12 Mb, consists of two

circular chromosomes, a characteristic feature of members of the *Vibrio* genus. Both chromosomal sequences were downloaded and processed for further analysis.

Prior to downstream applications, the retrieved genome sequence underwent a series of quality assessment steps to ensure data integrity and suitability for gene prediction. The FASTA file was examined for completeness, including verification of sequence length, absence of ambiguous nucleotide residues (such as excessive 'N' bases), and consistency in formatting. Particular attention was given to ensuring that the chromosomal contigs were complete and not fragmented, as incomplete assemblies may lead to inaccurate gene prediction results.

Sequence integrity was further validated by confirming the continuity of open reading frames (ORFs) and checking for the absence of assembly gaps or sequencing errors that could interfere with computational annotation. Basic quality checks were performed using standard bioinformatics tools to assess nucleotide composition, GC content distribution, and overall sequence stability.In addition, metadata associated with the genome, including strain information, accession number, and annotation status, were carefully reviewed to ensure that the selected dataset corresponded to a well-characterized and widely accepted reference strain. This step was essential to maintain consistency and reproducibility of the analysis.

Following validation, the curated genome sequence was used as the input for subsequent gene prediction and antimicrobial resistance analysis workflows.

4.3 Gene Prediction Using Prodigal

Protein-coding gene prediction was performed using Prodigal (version 2.6.3) in single-genome mode. The concatenated FASTA sequence of the V. vulnificus genome was provided as input. Prodigal was configured to output two principal files: a nucleotide FASTA file containing predicted gene sequences (genes.fna) and a protein FASTA file containing the translated amino acid sequences of all predicted coding sequences (proteins.faa). The self-training mode was employed to derive organism-specific codon usage and translation initiation models directly from the genome sequence, maximizing prediction accuracy for this organism without requiring external reference data.

4.4 AMR Gene Identification Using CARD and RGI

Predicted protein sequences were uploaded to the Resistance Gene Identifier (RGI) web interface and queried against the Comprehensive Antibiotic Resistance Database (CARD) using the 'perfect and strict hits' criteria. This stringency threshold was selected to minimize false-positive predictions while retaining high-confidence resistance gene identifications. The analysis reported resistance mechanism classification, drug class associations, and percent

identity to CARD reference sequences for each hit. Genes classified under the 'perfect' and 'strict' RGI confidence tiers were retained for further structural and docking analysis.

4.5 Target Protein Structure Selection and Preparation

For each identified AMR gene, the corresponding protein structure was retrieved from the RCSB Protein Data Bank (PDB) where an experimentally determined structure was available. In cases where no experimental structure was deposited, models generated by AlphaFold2 and available through the UniProt AlphaFold Structure Database were employed. All retrieved structures were inspected for completeness of the polypeptide chain, resolution quality (for crystallographic structures), and absence of significant missing loop regions proximal to putative active sites. Water molecules and co-crystallized ligands not relevant to the analysis were removed during structure preparation.

4.6 Ligand Selection and Preparation

Manoalide (PubChem CID: 6437368) was selected as the ligand molecule for this study following a systematic search of the Comprehensive Marine Natural Products Database (CMNPD), with emphasis on compounds derived from marine sponges that exhibit reported antimicrobial or enzyme-inhibitory activity. The selection criteria included structural uniqueness, documented biological activity, and potential relevance in targeting antimicrobial resistance mechanisms. Manoalide, a sesterterpenoid compound isolated from marine sponge species, satisfied these criteria and was therefore chosen for further computational evaluation.The three-dimensional (3D) structure of Manoalide was retrieved from the PubChem database in Structure Data File (SDF) format, which contains detailed atomic coordinate information required for molecular modeling. In addition to the structural file, the canonical SMILES (Simplified Molecular Input Line Entry System) string of the compound was recorded and utilized for subsequent pharmacokinetic and drug-likeness analysis using tools such as SwissADME.Prior to molecular docking, ligand preparation was carried out to ensure structural accuracy and compatibility with docking software. The retrieved SDF file was converted into Protein Data Bank (PDB) format using OpenBabel, an open-source chemical file conversion tool widely used in computational chemistry workflows. During this conversion process, hydrogen atoms were added where necessary, and proper bond orders were assigned to maintain chemical correctness.Subsequently, energy minimization was performed on the ligand structure to optimize its three-dimensional geometry and achieve a stable conformational state. This step reduces steric clashes and unrealistic bond angles, thereby improving the reliability of docking simulations. The minimized ligand structure was then saved in PDB format and used as the input for molecular docking studies.This systematic preparation of the ligand ensured that the molecular structure used in docking simulations was both chemically accurate and energetically favorable, thereby enhancing the reliability of protein–ligand interaction predictions.

4.7 Molecular Docking Using CB-Dock2

Molecular docking was performed using the CB-Dock2 web server. For each of the five AMR target proteins, the prepared receptor structure (PDB file) and the Manoalide ligand (PDB file) were uploaded to the server. CB-Dock2 was configured for blind docking: the CavityPlus algorithm first identified all geometrically feasible binding cavities on the protein surface, and AutoDock Vina was subsequently applied to perform docking calculations in each detected cavity. The binding energy (kcal/mol) and docking pose were recorded for the top-ranked cavity for each target protein.

4.8 Visualization Using PyMOL

Docking poses were downloaded from CB-Dock2 and imported into PyMOL (version 2.5) for structural visualization and interaction analysis. The protein–ligand complex was rendered to identify hydrogen bond donors and acceptors, hydrophobic contact residues, and van der Waals interactions within 4 Å of the docked ligand. Interaction diagrams were generated using PyMOL's built-in analysis tools and supplemented with two-dimensional interaction maps where applicable. Images were exported at 300 dpi for inclusion in this report.

4.9 ADME Prediction Using SwissADME

The drug-likeness and pharmacokinetic properties of Manoalide were evaluated using the SwissADME web server, a widely used online tool for predicting how a compound behaves in the human body. The canonical SMILES string of Manoalide was used as the input, as it provides a simple and standardized representation of the molecular structure for computational analysis. Upon submission, the server generated a range of important parameters that help assess the suitability of the compound as a potential drug, including the Lipinski Rule of Five criteria such as molecular weight, lipophilicity (logP), number of hydrogen bond donors, and number of hydrogen bond acceptors, which are commonly used to predict oral bioavailability. In addition, the analysis provided information on topological polar surface area (TPSA), which is associated with the compound’s ability to cross biological membranes, as well as predictions of gastrointestinal (GI) absorption to estimate how effectively the compound could be absorbed in the digestive system. The ability of Manoalide to cross the blood–brain barrier (BBB) was also evaluated to understand its potential interaction with the central nervous system. Furthermore, the study assessed whether Manoalide acts as a substrate for P-glycoprotein (P-gp), a transporter protein involved in drug efflux, and predicted its potential to inhibit key cytochrome P450 (CYP) enzymes, which play a major role in drug metabolism. Finally, a bioavailability score was generated to provide an overall indication of the compound’s suitability as an orally active drug, and all these parameters were considered collectively to evaluate the pharmacokinetic profile of Manoalide.

CHAPTER 5

RESULTS

5.1 Gene Prediction and Genome Analysis

Analysis of the Vibrio vulnificus genome (approximately 5.12 Mb) through the Prodigal gene-prediction pipeline yielded 4,352 putative protein-coding sequences. Screening of these predicted proteins against the CARD database using RGI with 'perfect and strict' criteria identified five high-confidence AMR-associated genes: varG, adeF, CRP, PBP3, and parE. The characteristics and resistance mechanisms of each gene are summarized in Table 5.1.

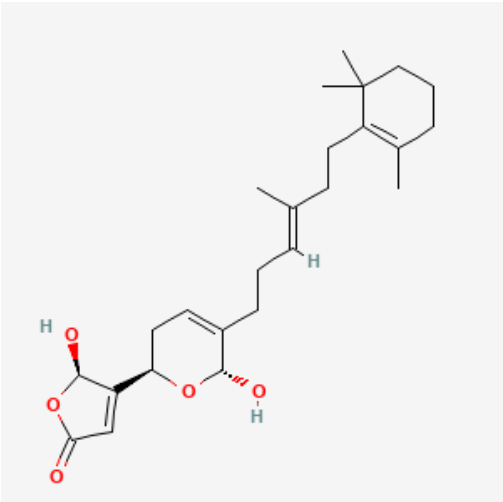


Figure 5.1: CARD-RGI output showing identified AMR genes in Vibrio vulnificus

Table 5.1: AMR genes identified by CARD-RGI analysis in Vibrio vulnificus

Gene	Function	Resistance Mechanism	Drug Class Affected
varG	Metallo-beta-lactamase	Antibiotic Hydrolysis	Beta-lactams
adeF	RND Efflux Pump	Active Drug Efflux	Fluoroquinolones, Tetracyclines
CRP	cAMP Receptor Protein	Transcriptional Regulation	Multiple Classes
PBP3	Penicillin-Binding Protein 3	Target Modification	Beta-lactams
parE	DNA Topoisomerase IV	Target Modification	Fluoroquinolones

5.2 Molecular Docking Analysis of AMR Proteins

Molecular docking of Manoalide against all five AMR target proteins was performed using CB-Dock2 with blind docking. The binding energies and best cavity assignments for each protein are presented in Table 5.2 and discussed individually below.

5.2.1 parE (Escherichia coli)

Best Binding Energy: −11.2 kcal/mol (Cavity 1). The most favorable docking result was observed with the parE subunit of DNA topoisomerase IV. The predicted binding geometry positions Manoalide within the ATP-binding pocket of parE, a site targeted by fluoroquinolone antibiotics. This complementary engagement with the enzyme's active site, with potential to disrupt the catalytic cycle underlying fluoroquinolone resistance and DNA replication, was clearly visualized in the docking output.

CurPocket ID	Vina score	Cavity volume (Å^3)	Center (x, y, z)	Docking size (x, y, z)	Contact residues	
⊙C3	-11.2	539	25, 68, 15	22, 22, 22	View	Li P
○C2	-8.7	599	16, 55, -13	22, 22, 22	View	Li P
○C4	-7.5	412	49, 65, 19	22, 22, 22	View	Li P
○C5	-7.2	384	17, 42, 5	22, 22, 22	View	Li P
○C1	-6.7	5704	34, 53, 2	22, 34, 35	View	Li P

Figure 5.2: CB-Dock2 docking result and PyMOL visualization — parE (E. coli)

5.2.2 Vibrio cholerae varG

Best Binding Energy: −9.5 kcal/mol (Cavity 5). Manoalide also displayed strong affinity for varG. The docking pose positions the ligand within the enzyme's catalytic cleft in a manner that could sterically obstruct substrate access and thereby impair beta-lactam hydrolysis.

CurPocket ID	Vina score	Cavity volume (Å^3)	Center (x, y, z)	Docking size (x, y, z)	Contact residues	
⊙C5	-9.5	327	1, 5, -7	22, 22, 22	View	L P
○C1	-7.4	1694	7, -17, -10	28, 22, 22	View	L P
○C2	-7.1	1016	-4, -16, -8	22, 22, 22	View	L P
○C4	-6.4	467	-3, -6, -13	22, 22, 22	View	L P
○C3	-6.1	594	-14, -8, 14	22, 22, 22	View	L P

Figure 5.3: CB-Dock2 docking result and PyMOL visualization — Vibrio cholerae varG

5.2.3 PBP3 (Haemophilus influenzae)

Best Binding Energy: −9.2 kcal/mol (Cavity 2). The docking results showed that Manoalide binds very well to the predicted cavity of the PBP3 protein. This interaction indicates that Manoalide may hinder the active site of PBP3, possibly obstructing cell wall biosynthesis.

CurPocket ID	Vina score	Cavity volume (Å^3)	Center (x, y, z)	Docking size (x, y, z)	Contact residues
⊙C2	-9.2	870	2, -3, 16	22, 22, 22	View
○C3	-8.0	622	9, -30, -22	22, 22, 22	View
○C1	-6.4	1026	8, -35, -32	22, 22, 22	View
○C4	-5.4	264	-8, -3, 4	22, 22, 22	View
○C5	-5.1	235	11, -18, -7	22, 22, 22	View

Figure 5.4: CB-Dock2 docking result and PyMOL visualization — PBP3 (H. influenzae)

5.2.4 CRP

Best Binding Energy: −9.1 kcal/mol (Cavity 1). A comparable binding affinity was observed with the CRP regulatory protein, raising the possibility that Manoalide may perturb the transcriptional networks that modulate resistance gene expression.

CurPocket ID	Vina score	Cavity volume (Å^3)	Center (x, y, z)	Docking size (x, y, z)	Contact residues
⊙C1	-9.1	11255	38, 54, 6	35, 35, 35	View
○C3	-6.9	1881	45, 52, -7	28, 22, 22	View
○C5	-5.8	572	17, 61, -10	22, 22, 22	View
○C4	-5.1	1782	62, 50, 0	22, 29, 33	View
○C2	-4.6	2376	14, 66, -25	22, 22, 22	View

Figure 5.5: CB-Dock2 docking result and PyMOL visualization — CRP

5.2.5 adeF

Best Binding Energy: −7.6 kcal/mol (Cavity 5). Binding to adeF was the least favorable among the five targets, though the moderate affinity recorded is nevertheless suggestive of partial occlusion of the efflux channel pore, which could reduce drug expulsion efficiency.

CurPocket ID	Vina score	Cavity volume (Å^3)	Center (x, y, z)	Docking size (x, y, z)	Contact residues
C5	-7.6	155	-6, -6, 12	22, 22, 22	View
C3	-6.7	199	14, 2, -41	22, 22, 22	View
C2	-6.0	218	6, 15, -26	22, 22, 22	View
C1	-5.8	221	2, -4, -17	22, 22, 22	View
C4	-5.6	166	-13, -11, 17	22, 22, 22	View

Figure 5.6: CB-Dock2 docking result and PyMOL visualization — adeF

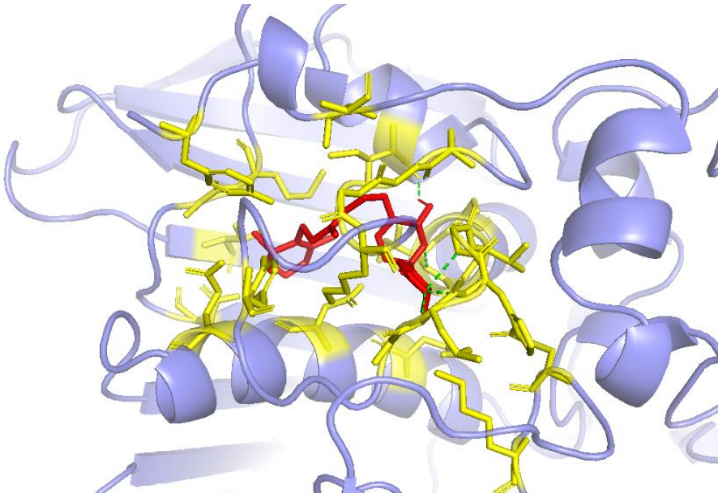
Table 5.2: Summary of molecular docking results — Binding energies of Manoalide with AMR proteins

Protein	Best Binding Energy (kcal/mol)	Cavity	Interaction Strength
parE (E. coli)	−11.2	Cavity 1	Very Strong
varG (V. cholerae)	−9.5	Cavity 5	Strong
PBP3 (H. influenzae)	−9.2	Cavity 2	Strong
CRP	−9.1	Cavity 1	Strong
adeF	−7.6	Cavity 5	Moderate

5.3 Protein–Ligand Interaction Analysis

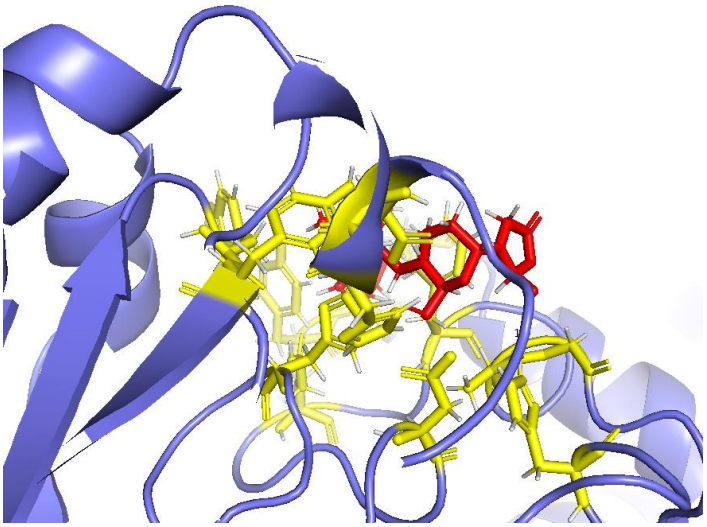
5.3.1 Interaction with parE (Escherichia coli)

Within the parE active site, Manoalide participates in an extensive interaction network. Seven hydrogen bonds were formed with GLN2037, GLU2038, ASP2041, ASN2042, SER2043, GLU2046, and THR2163, while hydrophobic contacts with VAL2039, VAL2067, LEU2091, ILE1010, and VAL2114 contribute non-polar stabilization. Additional van der Waals contacts with TYR1005, ARG2093, TYR2105, HIS2112, GLY2097, and GLY2098 reinforce the binding conformation.



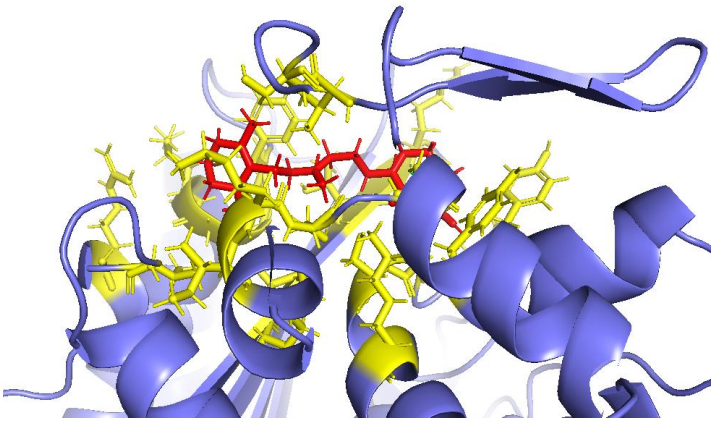
5.3.2 Interaction with Vibrio cholerae varG

In the varG complex, Manoalide is nestled deep within the active-site cavity. Polar contacts with GLU, ASN, SER, and THR residues anchor the ligand, while hydrophobic embedding within the VAL, LEU, and ILE-lined pocket confers additional stability. This positioning is consistent with competitive inhibition of the beta-lactamase catalytic mechanism.



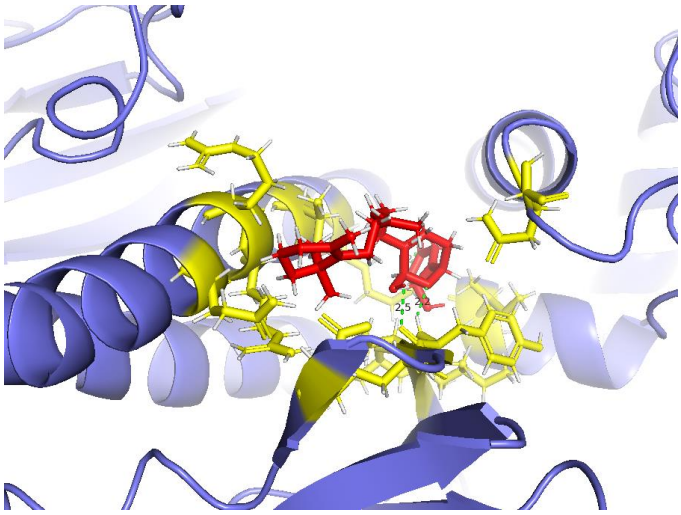
5.3.3 Interaction with PBP3 (Haemophilus influenzae)

The Manoalide–PBP3 complex revealed satisfactory steric and chemical complementarity. Hydrogen bonds with SER, THR, and GLN residues orient and anchor the ligand, while LEU, VAL, and ALA-mediated hydrophobic contacts consolidate the binding geometry.



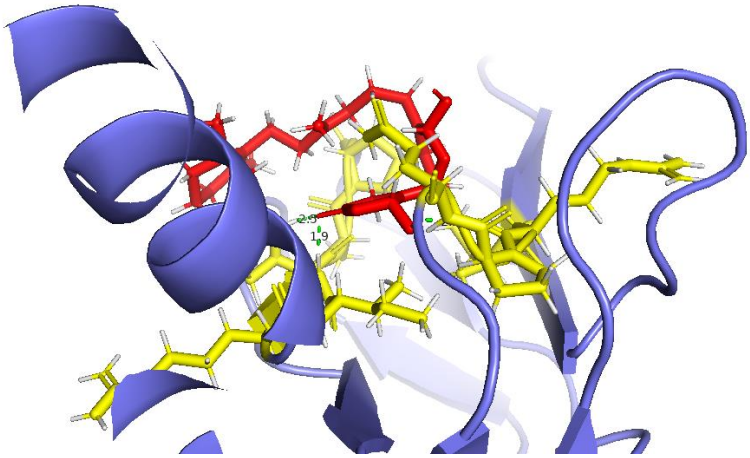
5.3.4 Interaction with CRP

Within the CRP binding region, Manoalide forms hydrogen bonds with ASP, GLU, and ASN residues that enforce a specific ligand orientation. Hydrophobic packing with VAL, LEU, and ILE residues provides the non-polar component necessary for thermodynamic stability of the complex.



5.3.5 Interaction with adeF

Even for the lowest-affinity target, adeF, the Manoalide complex adopted a geometrically plausible binding mode. Hydrogen bonds with GLN, SER, and THR stabilize the ligand position, while LEU, VAL, and PHE-mediated hydrophobic interactions provide non-polar reinforcement.



5.4 ADME and Drug-Likeness Analysis

Manoalide's drug-likeness and predicted pharmacokinetic behavior were comprehensively assessed through SwissADME. The principal parameters are consolidated in Table 5.3.

Table 5.3: SwissADME pharmacokinetic parameters of Manoalide and Lipinski's Rule of Five

Parameter	Manoalide Value	Lipinski's Limit	Status
Molecular Weight	416.55 g/mol	≤ 500 g/mol	✓ Pass
LogP (Lipophilicity)	4.07	≤ 5	✓ Pass
Hydrogen Bond Donors (HBD)	1	≤ 5	✓ Pass
Hydrogen Bond Acceptors (HBA)	4	≤ 10	✓ Pass
Topological Polar Surface Area (TPSA)	~63 Å²	—	Favorable
Gastrointestinal Absorption	High	—	✓ Favorable
Bioavailability Score	0.55	—	Favorable
Lipinski Rule of Five Violations	0	—	✓ Passed

Manoalide satisfied all four Lipinski criteria without exception. Its molecular weight (416.55 g/mol), logP (4.07), hydrogen-bond donor count (1), and hydrogen-bond acceptor count (4) fall comfortably within permissible ranges. The prediction of high GI absorption reinforces its candidacy as an orally deliverable therapeutic.

Figure 5.7: SwissADME analysis output for Manoalide

CHAPTER 6

DISCUSSION

The present study was designed to characterize the AMR landscape of V. vulnificus and, through an integrated computational framework, evaluate Manoalide as a candidate inhibitor. By sequentially combining genome retrieval, gene prediction, resistance gene identification, structure-based docking, interaction profiling, and ADME analysis, a coherent mechanistic picture of resistance and inhibition potential was assembled.

Prodigal's annotation of the 5.12 Mb genome yielded 4,352 protein-coding predictions, a figure reflective of V. vulnificus's considerable metabolic versatility. The organism's documented capacity to thrive across a range of temperature, salinity, and nutrient regimes implies a correspondingly broad functional gene repertoire, consistent with this gene count.

The five CARD-identified AMR genes illuminate the mechanistic breadth of V. vulnificus resistance. VarG encodes a metallo-beta-lactamase that destroys the beta-lactam ring of penicillins and cephalosporins. AdeF forms part of an RND-family efflux system that expels structurally diverse antibiotics including fluoroquinolones and tetracyclines. CRP, though principally a metabolic regulator, modulates the expression of numerous resistance-associated operons and thereby exerts indirect control over antibiotic susceptibility. PBP3's reduced affinity for beta-lactams confers insensitivity to this antibiotic class through target modification, while parE mutation renders DNA gyrase and topoisomerase IV less susceptible to fluoroquinolone inhibition.

The docking results demonstrate that Manoalide can productively engage all five AMR targets, positioning it as a potential multi-target agent. Its strongest affinity for parE (−11.2 kcal/mol) implies particularly favorable geometric and energetic complementarity with that enzyme's active site. Disruption of parE catalytic function would impair chromosomal decatenation during replication, with broad bactericidal implications independent of its role in fluoroquinolone resistance.

Strong binding to varG, PBP3, and CRP suggests that Manoalide's antimicrobial action extends across mechanistically distinct targets. VarG inhibition could re-sensitize the organism to beta-lactam antibiotics currently neutralized by enzymatic hydrolysis. PBP3 occupancy would compromise peptidoglycan cross-linking, potentially destabilizing the cell wall and increasing susceptibility to osmotic lysis. CRP perturbation may have regulatory cascade effects on resistance gene expression. The more modest adeF affinity still represents a potentially clinically meaningful partial efflux impairment.

Interaction analysis at atomic resolution corroborated the docking affinity data. Across all five targets, Manoalide formed an appreciable number of hydrogen bonds that constrain ligand orientation and confer specificity, complemented by hydrophobic packing interactions that provide the thermodynamic driving force for complex formation. The consistency of these contact patterns across targets suggests that Manoalide's chemical scaffold possesses inherent multi-target binding capacity.

The SwissADME pharmacokinetic assessment adds a critical translational dimension to the docking findings. All Lipinski parameters fall within permissible bounds, and the predicted high gastrointestinal absorption probability suggests Manoalide could achieve adequate systemic exposure following oral dosing — a prerequisite for clinical utility.

The present findings reinforce the proposition that marine natural product chemistry represents a productive frontier for antimicrobial drug discovery. Manoalide's structurally distinctive sesterterpenoid framework and multi-target binding capability offer a strategic advantage over mono-target agents: simultaneous inhibition of multiple resistance mechanisms raises the mutation barrier for resistance emergence, since spontaneous inactivation of several independent resistance pathways in a single organism is statistically improbable.

This study shows that using computational approaches to find possible inhibitors of antimicrobial resistance is effective. Genome analysis, identification of resistance genes, molecular docking, protein–ligand interaction analysis, and ADME evaluation collectively expedite the discovery of novel pharmaceuticals. Nonetheless, further experiments, both in vitro and in vivo, are necessary to confirm the compound's ability to inhibit, its toxicity, and its effectiveness as a therapeutic agent before human application.

CHAPTER 7

CONCLUSION AND FUTURE PROSPECTS

7.1 Conclusion

This study employed a fully computational, multi-stage approach to dissect AMR in V. vulnificus and evaluate the therapeutic potential of the marine sesterterpenoid Manoalide. Prodigal-based genome annotation of the 5.12 Mb genome generated 4,352 protein predictions, of which CARD-RGI screening identified five high-confidence AMR genes — varG, adeF, CRP, PBP3, and parE — collectively encoding a mechanistically diverse multi-drug resistance profile.

Structural models derived from PDB deposits and AlphaFold2 predictions served as receptors for docking studies. Manoalide (PubChem CID: 6437368) demonstrated productive binding to all five targets, with the strongest affinity observed against parE (−11.2 kcal/mol). Appreciable binding energies against varG, PBP3, CRP, and adeF further substantiate Manoalide's multi-target inhibitory profile.

Residue-level interaction mapping confirmed that each binding event is stabilized by a complementary combination of hydrogen bonds and hydrophobic contacts, lending structural plausibility to the docking-predicted affinities and supporting Manoalide's candidacy as a broad-spectrum AMR inhibitor.

SwissADME pharmacokinetic profiling confirmed Manoalide's compliance with Lipinski's rule of five and predicted high gastrointestinal absorption, indicating favorable oral bioavailability prospects.

In sum, the computational evidence presented here positions Manoalide as a promising lead compound against V. vulnificus AMR targets and provides a template for applying integrative bioinformatic workflows to the discovery of novel antimicrobials against multi-drug resistant marine pathogens.

7.2 Future Prospects

Looking ahead, in vitro enzyme inhibition assays and bacterial growth inhibition experiments are the immediate priorities to validate the computational predictions. Medicinal chemistry optimization of the Manoalide scaffold — guided by the residue-level interaction maps generated here — could yield analogs with improved affinity and narrower off-target profiles. Parallel screening of structurally related marine sesterterpenoids may uncover additional leads. Comprehensive toxicological characterization, including cytotoxicity panels and animal pharmacokinetics, will be essential prerequisites before clinical evaluation. Finally,

incorporation of machine learning models trained on validated protein–ligand interaction data could further accelerate candidate prioritization in future screening campaigns.

CHAPTER 8

APPENDICES

Appendix A: Software Tools and Databases Used

This study made use of several bioinformatics tools and publicly available databases to carry out genome analysis, identification of antimicrobial resistance genes, molecular docking, and pharmacokinetic evaluation.

Tool / Database	Purpose	URL / Reference
NCBI Genome Database	Retrieval of V. vulnificus genome sequence in FASTA format	https://www.ncbi.nlm.nih.gov/genome/
Prodigal v2.6.3	Prokaryotic gene prediction and protein sequence generation	Hyatt et al., 2010
CARD (Comprehensive Antibiotic Resistance Database)	Identification of AMR genes and resistance mechanisms	https://card.mcmaster.ca/
RGI (Resistance Gene Identifier)	Query protein sequences against CARD database	Alcock et al., 2022
CB-Dock2	Blind molecular docking with cavity detection	Liu et al., 2022
PyMOL v2.5	Visualization of protein–ligand interactions	Schrödinger LLC
SwissADME	ADME/pharmacokinetic and drug-likeness prediction	Daina et al., 2017
PubChem	Retrieval of Manoalide 3D structure (CID: 6437368)	https://pubchem.ncbi.nlm.nih.gov/
CMNPD	Comprehensive Marine Natural Products Database	https://cmnpd.org/

Appendix B: Parameters Used in Computational Analysis

Gene Prediction (Prodigal)

Gene prediction was carried out using default settings in single genome mode. The combined FASTA file (~5.12 Mb) was used as input, and the output included nucleotide sequences (genes.fna) and protein sequences (proteins.faa).

RGI Analysis

The predicted protein sequences were analyzed using the RGI tool with 'perfect and strict hits' criteria to ensure reliable identification of resistance genes. The latest version of the CARD database was used.

Molecular Docking (CB-Dock2)

Docking was performed using CB-Dock2 with AutoDock Vina as the docking engine. Blind docking was carried out, and the results included binding energies (in kcal/mol) along with different docking poses.

SwissADME

The canonical SMILES of Manoalide was used as input. The analysis included Lipinski's Rule of Five, gastrointestinal absorption, and bioavailability score.

Appendix C: Chemical Information of Manoalide

Property	Value
Compound Name	Manoalide
PubChem CID	6437368
Molecular Formula	$C_{25}H_{36}O_5$
Molecular Weight	416.55 g/mol
LogP	4.07
Hydrogen Bond Donors	1
Hydrogen Bond Acceptors	4
TPSA	~63 $Å^2$
Bioavailability Score	0.55
Source Organism	Luffariella variabilis (marine sponge)
Drug Class	Sesterterpenoid

Appendix D: List of Identified AMR Genes

Gene	Function	Resistance Mechanism	Drug Class
varG	Metallo-beta-lactamase	Antibiotic degradation (hydrolysis)	Beta-lactams
adeF	RND family efflux pump	Drug expulsion / efflux	Fluoroquinolones, Tetracyclines
CRP	cAMP receptor / regulatory protein	Gene expression regulation	Multiple
PBP3	Penicillin-binding protein 3	Target modification	Beta-lactams
parE	DNA topoisomerase IV subunit B	Fluoroquinolone target modification	Fluoroquinolones

Appendix E: Docking Summary (Detailed)

Protein	Binding Energy (kcal/mol)	Best Cavity	Key Interaction Types	Predicted Effect
parE (E. coli)	−11.2	Cavity 1	7 H-bonds; hydrophobic (VAL, LEU, ILE); van der Waals	Disruption of fluoroquinolone resistance; inhibition of DNA decatenation
varG (V. cholerae)	−9.5	Cavity 5	Polar contacts (GLU, ASN, SER, THR); hydrophobic embedding	Steric obstruction of beta-lactamase catalytic site
PBP3 (H. influenzae)	−9.2	Cavity 2	H-bonds (SER, THR, GLN); hydrophobic (LEU, VAL, ALA)	Inhibition of peptidoglycan cross-linking
CRP	−9.1	Cavity 1	H-bonds (ASP, GLU, ASN); hydrophobic (VAL, LEU, ILE)	Perturbation of resistance gene transcriptional regulation
adeF	−7.6	Cavity 5	H-bonds (GLN, SER, THR); hydrophobic (LEU, VAL, PHE)	Partial occlusion of efflux channel pore